PAPER-II

(GROUP-A)

RENAL SYSTEM & BODY FLUID

Renal System

1. Mention/List the functions of kidney. **(CMU:** Nov-22, May-21, Nov-21, 20, 19) (Ja-08,15,Ju-13, May-16,18,Ja-17,Nov-18) Function of PCT **(CMU:** May-23, Nov-23)
2. Enumerate the hormones that act on kidney with their site of action and function & their action **(CMU:** Nov-24, 22, 21) (Ju-12, 14, Nov-15). Briefly discuss the role (Nov-18) of kidney in regulation of blood pressure (Ja-09, 14, Nov-15), Role of kidney in H_2O Balance? (May-16)
3. Define GFR. Describe how it can be measured and list the major factors affecting it? **(CMU: May-25,** 23, **Nov-25**, 22, 21, 20) (Ja-11,12,14,15, Ju-07,14, Ju-15, Ja-16, 17, May-19,18)
4. Draw and lable the glomerular filtrating membrane **(CMU:** May-24) and explain the mechanism of filtration in this bed? (Ju-05, 13, May:15, Ju-15)
5. List the renal function test. **(CMU:** May: 22,Nov-20, 19) (Ja-10,13 May-15, Ju-15, Nov-16, May-19) Mention their clinical significance of physical tests (May-15, Ju-15, (Ja-05, 09, 10, 11, 12, 14 Ju-05, 08,14) **(CMU:**May-22, Nov-19)
6. Discuss in brief the peculiarities of renal circulation? **(CMU:** Nov-19) (Nov-16) How auto regulation is maintained in renal bed. (Ja-06, Ju-09, May-15)
7. Write down the characteristics of renal circulation. What are the factors responsible for auto regulation of renal circulation (Ja-13, Ju-13) write down special features of renal circulation. **(CMU:** Nov-24)
8. Define the auto regulation of renal circulation (Ju-13) and describe the mechanism of auto regulation. (Ju-05, 06,12 Ja-10, Ju-15)
9. Discuss the basic process of urine formation **(CMU:** May-23, 21, Nov-20, Nov-19, 18) (Ja-16, 17, May-16), Discuss the mechanism of formation of concentrated (Ja-07, Ju-09) and dilute urine. **(CMU:** May-22, Nov-21) (Ju-07, Ja-15, Nov-15, 18) Discuss the role of loop of henle to producing concentrated urine. **(CMU:** May-24)
10. Mention the mechanism of secretion of substance that are reabsorbed & secreted from PCT & DCT. (Ja-11, 12)
11. State the mechanism of acidification of urine. **(CMU:** May-23, 22) (Ju-06, Ja-12,13)
12. Explain why normal urine is acidic. (Ja-08)
13. Give the composition of normal urine. (Ja-08)
14. Name the endocrine substances that are secreted from the renal tubules?(Ja-07, Ju-09,14)
15. Name the different parts of renal tubules with their functions (Ja-13)
16. Calculate effective filtration pressure in the glomerular membrane. (Ju-08)
17. Define the counter current system (Ja-16). Discuss the role of loop of Henly in the formation of hyperosmolar medullary interstitium. (Ja-08) what is the highest osmolarity of renal medulla? **(CMU:** May-22)
18. How counter current mechanism in kidney operates to produce hypertonic/ concentrated urine. (Ja-10, Ju-11,13,14)
19. Explain the role of macula densa in autoregulation of GFR. **(CMU:**May-20) (Ja-11)
20. Role of ADH & aldosterore on Kidney **(CMU:** Nov-23)

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21. Mechanism of water reabsorption in different part of renal tabule **(CMU:** Nov-23)
22. Define counter current hypothesis (May-17) & How concentrated urine is formed. **(CMU:**Nov-22, May-20) what are two basic requirments to form concentrated urine? **(CMU:** May-22) Discuss the mechanism of formation of hypotonic urine.(Ja-16)
23. What do you mean by the plasma load and tubular load **(CMU:**Nov-19) explain with example.
24. What do you mean by free water clearance? (Ju-11) Explain the renal threshold for glucose is 180 mg/dl whereas TmG is 320 mg/min. (Ju-07, Ja-15, May-18)
25. Differentiate between tubular load and tubular transport maximum. Calculate the tubular load of glucose if plasma glucose level is 100 mg/dl and GFR is 120 ml/min. (Ju-09)
26. What is plasma clearance value (Ju-12)?
27. Illustrate its importance to evaluate renal functions. **(CMU:**May-20) (Ju-09)
28. What do you mean by glomerulotubular balance? Write down its importance. **(CMU:**May-20)
29. Explain how four starling forces affect to form glomerular filtrate. What is filtration fraction? (Ju-12) Calculate the Net filtration pressure. (May-16)
30. What will be the glucose clearance rate if tubular load is 150 mg/min. (Ju-11)
31. Describe the mechanism of renal tubular reabsorption of Na^+ and glucose. (Nov-15, Ja-16)
32. Determinants of GFR ? (Ja-17) Factors of GFR **(CMU:** Nov-22, May-22)
33. Briefly describe the Autoregulation of GFR (Nov-16,18, May-16)
34. What is renal threshold **(CMU:**Nov-22, 21, 19) & TmG? **(CMU:**Nov-22) & what is nephron (Nov-18,16)
35. Name the substances that are reabsorbed & secreted from PCT.**(CMU:**May-24) & DCT(May-17)
36. Briefly discuss about micturition relfex. **(CMU:** Nov-20)
37. What is nephron? Draw and label different part of nephron. **(CMU:**May-23, 21, 19, Nov-19) Classification of nephron. **(CMU:** Nov-24)
38. Name the hormones that regulate tubular reabsorbtion and write down their site of action and effects **(CMU:** May-19)
39. Difference between osmotic diuresis and water diuresis **(CMU:** Nov-19)
40. **Name the substance that are-**
    (a) filtered through glomerular membrane (b) reabsorbed from renal tubules and (c) secreated to the renal tubules. **(CMU:** May-19)
41. Discuss the kidney buffer system **(CMU**: May-21)
42. How GFR Auto regulated by the tubuloglomerular feedback system. **(CMU:** May-21)
43. Write down the importance of plasma clearance.**(CMU:** May-22)
44. Describe the factor affecting the filtration through glomerular filtration membrance.**(CMU:**May-24)
45. Define obligatory urine volume. **(CMU:** Nov-24)
46. **SN:** a) Plasma clearance value (Ja-06,07,10, Ju-11,14)
    b) Polyuria (Ju-06) c) Automatic bladder (Ja-08)
    d) TmG value (Ju-03) e) JG apparatus (Ju-14, Nov-17)
    f) Renal threshold (Ja-15, May-18) g) Micturition reflex (May-17)
    h) Renin i) Atonic bladder (May-17)
    j) Free water clearance (Nov-15, 18) k) JG cells (Ja-07, 10)
    l) ORS (Ju-09) m) Obligatory urine volume. **(CMU:** May-19) (Ju-09, Nov-15, May-17)

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47. Explain he role of JG appartusin the regulateion of RBF and glomaular siltration rate. **(CMU: May-25)**
48. Briefly discuss the mechanism of glucose reabsorption from renal tubule to blood. Why maximum reabsorption occurs in PCT. **(CMU: May-25)**
49. What are the role of kidney in regulation of ECF osmolority. **(CMU: Nov-25)**
50. How GFR is auto regulated. **(CMU: Nov-25)**

Body fluid

1. What is water balance? Name the various fluid compartments in the body with normal values (Ja-17) and give the principle of body fluid measurement? (Ja-05)
2. How you can measure the amount of fluid deficit in case of dehydration. (Ju-06)
3. Tabulate the H_2O input & output chart.
4. What is ECF & ICF? Give the constituents of body fluid.
5. What are the difference between ECF and ICF (Ja-17)
6. Function of calcium in our body.
7. Briefly discuss about cerebrospinal fluid. (Nov-18)
8. Name the importance of different electrolyte (May-19)
9. **Short notes:-**
   a) Negative water balance (Ju-09)
   b) Water balance
   c) Water intoxication
   d) Metabolic acidosis & alkalosis

ENDOCRINOLOGY

Introduction

1. Name the endocrine glands (Nov-16, Ja-17)
2. Classify hormone according to their mode of action. (Ja-17)
3 Describe the mechanism of action of hormone. **(CMU:** May-20) (Ja-06, Nov-15, 18) Describe one of them (Nov-18).
4. Hormones of Hypothalamus. Draw and level the communication between hypothalamus and hypophysis. **(CMU:** May-19)

Pituitary Hormone

1. Name the growth promoting hormone. Write in brief the physiological effects of growth hormone on bones and cartilage. (Ja-07, 10) **(CMU:** May-24)
   Name the hormones secreted from pituitary gland ? **(CMU:** Nov-24, May-21) (May-16)
2. Write down the mechanism of action of GH. (Ja-06)
3. What is somatomedin? Elaborately discuss the role of somatomedin on somatic growth. **(CMU:**May-20) (Ju-10,11) write the function of somatomedin. **(CMU:** Nov-24)
4. Name the hormones secreted from the pituitary gland? What is pituitary diabetes? (Ja-07)
5. How hypothalamus control pituitary secretion? (May-16) Mention the relation of pituitary gland with nypothalamus **(CMU:** Nov-24)
6. Write down the role of ADH in regulation of normal body fluid osmolarity (Nov-16, Ja-17)
7. What do you mean by diabetogenic & ketogenic effect of growth hormone? (May-17)
8. List the function of Growth hormone. **(CMU:** May-22), (May-19)
9. **S.N:** a) Acromegaly **(CMU:** Nov-20) (Ja-11,12,13, May-15, Nov.15)
   b) Dwarfism(Ju-07,12, 13) . c) Milk ejection reflex (Ju-12)
   d) Gigantism (Ju-15) e) Diabetes insipidus **(CMU:** Nov-22, May-20)

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Thyroid Hormone

1. Name the hormones **(CMU:** Nov-22) of thyroid gland (Ju-15). State the steps of biosynthesis of thyroid hormones **(CMU:** May-22, Nov-20, 19) (Ja-05,08, 09, 14, Ju-05,10, 13, Nov-15)
2. Name the thyroid function tests with clinical significances with interpretations. (Ja-17, Nov-18)
3. State the calorigenic action of thyroid hormone. Briefly discuss the physiological effect of thyroid hormones on five target tissue. (Ja-08, 11, May-15)
4. How thyroid hormone secretion is regulated? (Ju-08, May-15, Ju-15)
5. Write down the mechanism of action of thyroid hormone. List the function of thyroid hormone on CVS. **(CMU:** Nov-23, May-19)
6. Difference between pituitary dwarf and a cretin body (Nov-15)
7. List the functions of thyroxin (Ja-17, Nov-18)
8. Write down the feature of hypothyroidism. **(CMU:** May-20)
9. Different between myxedema and cretinism **(CMU:** Nov-20, 19)
10. **S/N: -** a) Cretinism (Ja-09,12, Ju-06,14)
    b) Myxoedema (Ju-10,12,14, Ja-16, May-17)
    c) Grave's disease (Ja-09, 14)
    d) Thyrotoxicosis **(CMU:** May-21) (May-16)
    e) Goiter (Ju-13)
11. Defference between dwarfism & cretinism **(CMU:** May-22)
12. Function of TSH **(CMU:** Nov-22)

Parathyroid Hormone

1. Name the hormones that regulate the blood calcium level **(CMU:**Nov-24, 23, 20) (Ju-12, Ja-16). What is tetany & calcium rigor? (Ja-05, Ju-14) Function of calcium in our body. **(CMU:** Nov-24)
2. Role of PTH in regulating blood Ca^{++} level **(CMU:** Nov-23) (Ju-12). Summarize the functions of Ca^{++} (on muscle & blood). **(CMU:**May-21) (Ja-07, Ju-14, May-16)
3. Write down the function of parathyroid hormone. (Ja-16)
4. Show in a diagram the hormonal control of Blood Ca^{2+} (May-16)
5. **S/N:** a) Tetany b) Hypercalcemia c) PTH (Ju-13)

Pancreatic Hormone

1. Name the hormones of islets of langerhans? What are the effects of insulin on adipose tissue and muscles? (Ju-09, 11) **(CMU:** May-24)
2. Write down the consequence of insulin deficiency. Difference between DM & DI. (Ja-11,12)
3. Name the hormones controlling the blood glucose level. Illustrate their mode of action.(Ju-07)
4. Mechanism of action of insulin? Enumerate the effects on insulin lack? **(CMU:** May-22)
6. **S/N:** a) Diabetes mellitus (Ja-15, May-18) b) Diabetes insipidus (May-18)
    c) OGTT d) Insulin **(CMU:** May-21, Nov-19) (May-15,16 Ju-15)
7. List the hormone increase blood glucose. How insulin decrease blood glucose level. **(CMU:** May-19) (Nov-16, May-19)
8. Difference between diabetes mellitus & Diabetes insipidus (May-17)
9. List the factor that increase insulin secretion. **(CMU:** Nov-19)

Adrenal Hormone

1. Name the hormones secreted form adrenal cortex? **(CMU:** May-23) (May:15, Ja-13,14,15)
2. Why aldosterone is termed as life saving hormone? **(CMU: May-25)** (Ju05-08, Ja-09,13, 14). Name the steroid hormones (Ju-13)
3. Name the hormone of adrenal gland. Summarize the functions of cortisol **(CMU:** Nov-21) (May-18).
4. Elaborate the function of any one of the adrenal hormones. **(CMU:** Nov-19) (Ju-06)

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5. What is primary & secondary aldosteronism? Summarize the function of cortisol. (Ju-07)
6. Explain the mechanism of action of aldosterone. What are the physiological benefits of increased secretion of cortisol in stressful situation of the body? (Ja-08, May-15)
7. Name the hormone of adrenal gland (May-19) and illustrate their role in maintaining blood volume. (Ju-09)
8. Enumerate the non metabolic role of cortisol (Ju-12). What is permissive action?
9. **SN:**
   a) Cushing syndrome. (Ju-05, Ja-14, Nov-16)
   b) Addison's disease (Ja-17)
   c) Aldosterone escape. (Ja-07, Ju-11)
   d) Adrenogenital syndrome. (Ja-08)
   e) Conn's syndrome. (Ja-09, Ju-15)
   f) Adrenal/ Addisonian crisis
   g) Catecholamine
   h) Aldosterone.
10. Write down the metabolic function of cortisol (Ja-16) Function of cortisol **(CMU: May-23)**
11. Discuss the role of cortisol in stress & inflammation **(CMU**: May-22) (Ju-06,07,11, 13, Ja-12,15, Ja-16)
12. Write down the feature of Acromegaly. **(CMU:** Nov-19)
13. Write down the anti-inslammatary role of cortisol. **(CMU: May-25)**
14. Write diwn the function and regulation of aldosterone. **(CMU: May-25)**
15. List the metabolic effects on insulin. **(CMU: Nov-25)**
16. Explain the thyroid hormone secration, regulation. **(CMU: Nov-25)**

Reproductive System

1. What is spermatogenesis? **(CMU:** Nov-23, Ja-17) Write in brief the steps of spermatogenesis. (Ja- 17, Nov-15) Give its hormonal regulation. (Ja-06, 07, 08, 12, Ju-11, May-15, Ju-15). With diagram **(CMU:Nov-23)** Factors regulate spermatogenesis **(CMU:** Nov-21, 20)
2. Name the gonadotropins. Discuss their role in female reproductive life. (Ja-06)
3. Show in a diagram hormonal regulation of menstrual cycle. (Ju-08,09,14, Ja-09,11,12,14, May-15, Ju-15)
4. What are phases of menstrual cycle with Diagram? **(CMU:** May-22, Nov-19)
5. Explain the mechanism of menstruation. **(CMU:** May-22,Nov-21) (Ju-07,Ja-13,15, Ja-16, May-18)
6. Define ovulation? **(CMU:** Nov-22, 21) (May-15,17,Ju-15, Ja-16)
7. How it is regulated? What is the role of contraception in this regulation? (Ja-05, Ju-06,14).
8. What are the indicators of ovulation? **(CMU:**Nov-20) (Ju-07, Ja-15, May-19)
9. Discuss the mechanism of ovulation **(CMU:**Nov-20) (May-16, 17)
10. Name the changing follicle in ovarian cycle. Show in a diagram the hormonal control for ovulation. **(CMU:** Nov-22) (Ju-07,08,09,15, Ja-09,15, May-18)
11. Hormones, which control the ovarian cycle. Features of anovulatory cycle. (Ja-10, Ju-08)
12. Write down the fate of corpus luteum. (Ju-06)
13. Define menarche and menopause. Explain the physiological basis of menopause. (Ja-08,10)
14. Write down the composition of menstrual blood. (Ja-11, 12)
15. Role of prolactin and oxytocin on breast. Why lactation initiates after delivery? **(CMU:** May-19) (Ju-10,11) *Function of prolactin* **(CMU:** May-22)

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16. Describe the physiological effects of progesterone. (Ju-12). Enumerate the secondary sexual characteristics of male & female.
17. What is puberty? Write down the changes in male during puberty. **(CMU:** Nov-19) (Nov-16, 18)
18. Explain the role of hormones acting on breast **(CMU:** May-23, 19) **(CMU:**Nov-22, 21)
19. Name the hormones of placenta? (Ja-14, May-19)
20. Write down the actions of testosterone in male? (Ja-15, Nov-15, May-19)
21. Name the source of male hormone? Write in brief the functions of this hormone. (Ju-14)
22. Name the hormone of male & female sex organ. **(CMU:**Nov-19) (Ja-15, Nov-15, May-18)
23. Describe feature of male and female at puberty? **(CMU:**May-19) (Ju-15, May-16)
24. Write down the function of testosterone **(CMU:**Nov-20) (Ja-16, May-17, 18)
25. What are the feature of mature sperm? (Ja-16)
26. Define female sexual cycle (Nov-18). Briefly discuss the role of different hormones on uterine cycle. (Nov-16)
27. Describe ovarian cycle (Nov-18)
28. What is safe period (May-19)
29. What is Semen? Write down its composition. **(CMU:**May-20)
30. List the function of projesterone. State the physiological changes that occur during menopause. **(CMU:**May-20)
31. What is LH surge? **(CMU:**Nov-20)
32. **SN:** a) Puberty (Ju-06,14, Ja-07, Ju-15, Ja-16)
    b) Oogenesis (Ja-13)
    c) LH surge(Ja-13)
    d) Testosteron **(CMU:** May-22)
    e) Estrogen **(CMU:** Nov-22)
    f) Anovulatory cycle. **(CMU:** May-22, Nov-20, 19) (Ja-11)
    g) Azoospermia. (Ju-07, Ja-15, May-18)
    h) Amenorrhea. **(CMU:**May-21) (May-15,16 Ju-15)
    i) HCG hormone (Ja-07, 14)
    j) Safe period (May-17)
    k) Menstruation (Ju-13, Ja-17)
    l) Milk let down process **(CMU:**Nov-21) (Ju-12)
    Milk ejection reflex **(CMU:** May-23, Nov-22)
    m) Spermatogenesis **(CMU:** May-21) (Ju-13,14, Nov-15, May-16)
    n) Overion cycle (Nov-15)
    o) Lactation (May-17, Nov-18)
    p) Semen **(CMU:** Nov-21, May-19)
    q) Fetoplacenta unit **(CMU:** May-19)
    r) Prolactin. **(CMU:** May-20)
33. Name the placental hormones. Write down their role in maintaining pregnancy. What is sterility. **(CMU: May-25)**
34. Write down the composition of semen. What is the effects of sperm count on fertility. **(CMU: Nov-25)**

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SAQ - PAPER-II

(GROUP-B)

NERVOUS SYSTEM

duction

Define and classify the neuron. **(CMU:** May-20, Nov-19) (Ja-15, May-18, 19) Differences between axon and dendrite. (Ja-17) & classify according to conduction velocity. **(CMU:**May-24)
List the various types of nerve fibers **(CMU:** Nov-24) (May-15) and mention their properties **(CMU:**May-22) (Ju-12, May-19). Explain any two of them. (Ja-05, 08, May-19) Function of nerve fiber. **(CMU:** May-24)
Classify nerve fibers on the basis of diameter **(CMU:** Nov-19) and velocity of conduction of impulse, Describe the pathophysiology of denervation hypersensitivity? (Ja-07,10, Ju-14)
Discuss the different methods of conduction of nerve impulse through nerve fibre.

5. Mention the properties of neuron. (Ja-15, May-18)
6. Mention the sensory tracts with their function **(CMU:** May-21, Nov-20)
7. Describe the sequences of event of synaptic transmission (Ja-16)
8. List the neurotransmitter **(CMU:** Nov-20, 19) (Nov-16, 18)
9. Compare between sympathetic & parasympathetic nervous system **(CMU:** Nov-23, 19, 16) (May-19)
10. Draw & label a neuron. **(CMU:** May-20)
11. Classify motor neuron with example. Differentiate between upper and lower motor neuron usion. **(CMU:** May-23, Nov-21)
12. **S/N-** a) Wallerian degeneration. (Ju-07).
    b) Neurotransmitter. (Ja-14, Nov-15) c) Nerve fiber (Ja-17)
13. Mention the functional classification of nervous system. **(CMU: May-25)**
14. Write down the effects of autonomic nervous system on eye, evsand GIT. **(CMU: May-25)**

Synapse

1. Define synapse **(CMU:**Nov-19). Write in brief properties of the synapse **(CMU:** May-20) & classify synapse? **(CMU:**Nov-22) (Ja-11,12,13, Nov-16, 18)
2. Draw and lable synaptic transmission and elaborate the mechanism of synaptic inhibition? (Ju-09) **(CMU: May-25)**
3. Define & classify the neurotransmitter? **(CMU:** Nov-24,23,15, May-21,16) (Ja-08,07,11,12,16)
   How does neurotransmitter secrete at synapse (Ju-15, May-16)
4. Discuss the synthesis, degradation & functions of any one of the neurotransmitters (Ja-13)
5. Define the muscle tone. Discuss the transmission of excitatory impulses through a chemical synapse. **(CMU:**Nov-19) (Ja-08)
6. Mechanism of transmission of impulse across a neuromuscular junction (Nov-15, 18)
7. What do yoy mean by synaptic delay & synaptic response. **(CMU:** May-24)
8. **Short note:**
   a) EPSP and IPSP **(CMU:** Nov-23, May-19) (Ju-12) b) Synaptic inhibition. (Ju-06, Ja-09)
   c) Synapse. **(CMU:** Nov-21) (Ja-15, 17, May-17, 18)
9. Draw and level chemichal synapse. **(CMU: Nov-25)**

Receptor

1. Define & classify receptor. **(CMU: May-25**, 24, Nov-20) (Ja-08, May-15, 16, Ju-15, Ja-16)
   Define & classify Sensory receptor with example (Ja-17) **(CMU:** Nov-23, May-22, 19)
   What is receptor potential.
   Differences between receptor potential & action potential **(CMU: May-24)**
2. Mention the properties of receptor (May-15, 16 Ju-15, Ja-16). Illustrate Webber Fecher's law.

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(Ju-07, Ja-16)

3. S/N- a) Receptor (Ja-14) b) Name some chemoreceptor. (Ju-14)
   c) Withdrawal reflex **(CMU:**May-20) d) Reflex **(CMU:** Nov-21)
4. Name the general senses with their reception. **(CMU: May-25)**
5. What is lebeled line principal. **(CMU: May-25)**

Reflex

1. Define reflex (May-19). With classification **(CMU:**Nov-24, 23) Name the different properties of reflex action **(CMU:**Nov-19) (Ju-13, Nov-15) and write in details any two of them. (Ju-05, Ja-11)
2. Show in a diagram the components of reflex arc. **(CMU:** Nov-19). (Nov-15, May-19). Write down the clinical importance of eliciting a reflex. (Ja-06,Ju-10, 11)
   Draw and label a reflex are. **(CMU:** Nov-24)
3. List the various types of reflex (Ju-06, 10, 13). Draw & lable a complete pathway of polysynaptic reflex (Ju-12)
4. Draw complete pathway of monosynaptic reflex. **(CMU:** May-22, Nov-19), (Nov-16) List the properties of reflex action. (Ja-09, 14, Ju-13)
5. **Short note:** a) Stretch reflex. (Ja-08, Ju-08)
   b) Reflex **(CMU:**Nov-22) (Ju-15) c) Withdrawal reflex **(CMU:** May-20)
6. Mention the clinical importance of monosynaptic reflex (Nov-16)
7. Draw & label the reflex arc. (May-17)
8. Explain that knee jerk is a refiex. **(CMU:** Nov-23, May-19)
9. Name the sensory and motor tract. (Nov-23, May-19) 10. Babiniski sign. **(CMU:** Nov-22)
11. Define and classify reflex with example. properties of reflex. **(CMU:**May-21)
12. Golgi tendon reflex is a protective mechanism. Explain. **(CMU: May-25)**

Tracts & Pathway

1. Define and classify sense (Ju-12) How crude touch sensation is carried from periphery to center. **(CMU:**May-20) (Ju-09)
2. Mention the effects of sympathetic & parasympathetic nerves on GIT & heart . (Ju-12)
3. What do you mean by fight & flight response. (Ju-11,12)
4. Describe pathway and function of tract of GALL and BURDACH.
5. What is masking effect? **(CMU:** Nov-19) Trace the pathway of hearing **(CMU:**Nov 19) cause of nerve deafness. (Ju-11, May-15, Ju-15)
6. Name the motor pathways. (Ja-11,12,17). Motor tract **(CMU:**Nov-24) What is motor unit (Ju-12)
7. Name the different sensory ascending tracts. **(CMU:** Nov-22) (Ja-11, Ja-12,15, Ju-15, May-16, 17, 18) Trace the fine touch sensation from periphery to center? (Ju-05,12, 13 Ja-11, 14, Ja-09, May-17). Trace the pathway of spinothalamic tract. (Ju-15, May-16)
8. Trace the pain pathway from periphery to center. **(CMU:** Nov-19) (Ja-05,07,08,11,12,15, Ju-06,14, Nov-15, 16, 18, May-18)
   Trace the dual pathway for trasmission of pain signal from periphery to central nervous system.**(CMU: May-25, 24, Nov-25)**
   What do you mean by referred pain **(CMU:** May-23, Nov-18)
9. State in brief the analgesia system in the body. **(CMU:May-25,** 23, 20) (Ja-07, 08)
   What is analgesia system in the brain & spinal cord. **(CMU:** May-24)
10. Explain why Babinski's sign is positive in pyramidal tract lesion. (Ja-05)
11. What are the features of spinal hemi-section at the level of L_2? (Ja-07, Ju-12,14)

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12. Trace pathway of fine touch sensation from periphery to center. What are the features of transection of spinal cord? (Ju-07) Trace the crude touch pathway **(CMU:** Nov-23)
13. Show in a diagram the structure of a muscle spindle. Explain the role of muscle spindle in maintaining muscle tone. (Ja-09, Ju-10) **(CMU:**May-19)
14. How pain pathway can be modulated? **(CMU:** May-22, Nov-22) (Ju-06)
15. Compare and contrast between the pyramidal and extra pyramidal lesion. **(CMU:** May-20) What do you mean by release phenomenon? (Ju-09)
16. Differentiate between pyramidal and extra pyramidal tracts. (Ju-06,Ja-06,11,12)
17. Compare & contrast the dorsal column & anterolateral system of sensory tract. (May-23) (Ju-10)
18. What is UMN & LMN. Differentiate between UMN & LMN lesion. (Ju-13, Ja-14)
19. **SN:** a) Referred pain. (Ja-10, Nov-15)
    b) Analgesia system (Ju-12)
    c) Brown sequard syndrome. **(CMU:** May-23, 19) (Ja-05, 08, 11, 12, Ja-16)
    d) Spinal Shock. (Ju-14)
    e) Bell magendie law. (Ju-14)
    f) Pyramidal tract (Ju-15 May-16)
20. [Name the motor neurons] **(CMU:** Nov-24, May-20) with their location (Ja-14, Ju-14, May-18)
21. Locate the motor neurons in CNS. Trace the pathway of pyramidal tract.**(CMU:** May-22) (Ja-15, May-18) Define pyramidal & extrapyramidal tract **(CMU:** May-22)
    Function of extrapyramidal pathway. **(CMU:** May-24)
22. Compare & contrast between the two division of autonomic nervous system. **(CMU:** Nov-22) (Ja-15)
23. Show in a flow diagram the motor hierarchy with their performance. (Ju-14)
24. Write down the feature of upper and lower motor neuron lesion (Ja-16) Defferentiate between UMNL & LMNL. **(CMU:** Nov-24, 22) Define UMN & LMN **(CMU:** Nov-24)
25. Origin, Course, termination and function of corticospinal tract **(CMU:**May-24, Nov-24)(Nov-17)
26. Name the primary sensation of taste. Trace the pathway of taste (Nov-16, May-16)
27. Name the sensory and motor tract (May-19)
28. Trace the pathway of spinothalamic tract **(CMU:** May-21)
29. Trace the pathway of crude touch from periphery to center. Name some ascending tract. **(CMU:** Nov-23, 21)
30. Name the sensory tracts occupy the posterior column of spinal cord. **(CMU: Nov-25)**

Cerebellum

1. Name the cerebellar nuclei and fibers. How does cerebellum perform the error **(CMU:** May-19) control mechanism. (Ju-06)
2. Explain how cerebellum corrects the errors of movement? (Ja-05, 06, May-15)
3. Differentiate the cerebellar and basal ganglia lesion. (Ja-08)
4. Draw and lable the neuronal circuit of the cerebellum. Discuss the role of cerebellum in the voluntary motor function. (Ja-07, Ju-10)
5. What is cerebellar ataxia? (Ju-05, 08, Ja-09, 11, 12, May-15)
6. List the function of cerebellum. **(CMU:** May-23, 20, 19, Nov-22, 21,16) (Ja- 11, 12, 14, 17, Ju-10, 13)
7. Give the effects/features of cerebellar lesion **(CMU:**Nov-22, 21, 19) (Ja-13, 14, 17 Ju-13, Nov-16, May-23,19)
8. Briefly discuss about basal ganglia. (Nov-15, May-17) & What is memory (May-17)
9. S/N: Cerebellum **(CMU:**May-22, 21) (May-16)
10. List the components and function of basal ganglia. **(CMU: May-25)**

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Hypothalamus

1. List the functions of hypothalamus.**(CMU:**May-24,Nov-20)(Ja-12,13, 14, 17, Ju-13, Nov-15, 18) Write its relation with satiety. (Ju-05, 06, 08, Ja-10)
2. Explain how hypothalamus controls the normal temperature in the body in hot (Ju-13, Nov-15,18) & cold climate. **(CMU:**Nov-20) (Ju-11, Ja-14, 17, Nov-15, Ju-15) How temperatures is gained and lost from our body? (CMU: May-23, Ju-15) What is hyperthermia **(CMU:** May-23)
3. What is satiety? (May-15). Compare and contrast between hunger and satiety. (Ju-08, Ja-11, 12)
4. Role of hypothalmus in body water regulation & body temparature regulation. **(CMU:** May-24)

Limbic System

1. List the components of limbic system (Ju-13,14, Ja-14, May-15, Ju-15). What is its relation with memory & emotion? (Ju-13,15 Ja-10, 14, 16, May-15,16)
2. Define memory. Write down the functions of limbic system. **(CMU:**Nov-23, 19), (Ja-06, 09, Ju-06,13,14, May-16)
3. Name the structure of the limbic system. (Ju-06) Explain how set point of body temperature control mechanism is maintained. (Ja-08)
4. Function of thalamus. **(CMU:**May-20)
5. **SN:** a) Sleep **(CMU:**May-20) (Ju-12, 13)b) Blood brain barrier. (Ja-08)
   c) Thalamic syndrome. (Ju-06,14) d) Memory. (Ju-10, Nov-16, May-19)
   e) Limbic system (Nov-16) f) CSF (CMU-Nov-19)
6. What is intention tremor? **(CMU: Nov-25)**

Special Sense

1. Draw and label the visual pathway **(CMU:**May-21) (Ja-13, Ju-13). What do you mean by homonymous hemianopia. (Ju-05, 09, Ja-10,13) function of visual receptors (May-17, Nov-18)
2. Name the reflexes related to the eye? Draw and lable **(CMU-**Nov-19) the pupillary light reflex pathway? **(CMU:** Nov-22, 21, May-21) (Ju-07,14, Ja-07,15, May15,18, Ju-15)
   Write down the importance of pupillary reflex pathway. (Ju-14)
3. Show in a diagram the pathway for vision & mention the defect results from lesion of optic chiasma.
4. What do you mean by light reflex? Write down the pathway of light reflex. **(CMU:**Nov-20) (Ja-08, Nov-15,18, May-16)
5. Name and locate the receptor for hearing. **(CMU:**Nov-22) (May-17, 18) Trace the pathway from this receptor to center. **(CMU:**Nov-22, 20) (Ju-07, Ja-15, Nov-18) What is near response ?
6. Give the mechanism of hearing. **(CMU:**May-21) Tests perform to detect deafness (Nov.15)/ Cause of nerve deafness. **(CMU:**May-21)
7. Mention the changes occur in accommodation for near **(CMU:**May-22) & distant vision. **(CMU:**Nov-20) (Ju-10, 11, Nov-16) Define accommodation **(CMU:** Nov-20)
8. Name the chemoreceptor for taste sensation. Trace the taste pathway from this receptor to center. **(CMU:** May-22) (Ju-07, 14, May-19) Name the primary sensation of taste **(CMU:** Nov-24, May-20) Trace the taste pathway **(CMU:** Nov-24)
9. What are the receptors of special sense and location? **(CMU:** May-22, Nov-19), (Ja-13) Name the special sense. (May-17, 19)

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10. Define & classify sense. (Ju-12, Nov-15)
11. What are the common errors of refraction? (Ja-14, 17 Ju-14) Discuss any two of them. (Ju-06, 12 Ja-05, 09, 10, 13) Discuss their causes & correction (Ja-14, 17, May-16)
12. Name the receptor of eye with their function. **(CMU:** Nov-22) (Ja-15, M-18)
13. Name the general senses. How does these sensation transmitted from peripheri to center (Ja-16)
14. Compare and contrast the short and long sighted vision. (Ja-16)
15 State the visual impairment in left optic nerve and left optic tract is cut **(CMU**: Nov-19)
16. Mention the common error of refraction with correction. **(CMU:**May-22, 21, 20)
17. Write in short about dark adaptation. **(CMU:**May-20)
18. What are the modalities of taste sensation? **(CMU:** May-22)
19. **SN:** a) Accommodation reflex **(CMU**: May-21, Nov-19) (Ju-05, 08, 13, Ja-11, May-15)
    b) Deafness (Ju-06, 07, 11, 13, Ja-12, Nov-15, Ja-16)
    c) Define classify and causes of Deafness (Ja-17, May-16, 19) **(CMU:** Nov-19)
    d) Dark adaptation (Ja-07)
    e) Rhodopsin **(CMU**-May-19) (Ju-11)
    f) Receptors of special sense(Ja-13)
    g) Near vision (Ju-10, 11)
    h) Sense (Nov-18)
    i) Refractory media (Ja-15, May-18)
    j) Myopia **(CMU:**Nov-20)
    k) Color vision **(CMU:**Nov-19) (Ja-07, Nov-15, May-19)
    l) Taste bud **(CMU:**Nov-20) (Ja-10, Ju-10, May-15, 17, Nov-18)
20. Trace the smell pathway. **(CMU: Nov-25)**

Heat & Temperature

1. Define body temperature. Site for measurement of body temperature.
2. How body temperature (May-17) is regulated in hot & humid climate **(CMU:**Nov-21)(May-18). What is hyperthermia **(CMU:**Nov-21)(May-18, Ja-07, 15, May-16)/hypothermia **(CMU:**Nov-22)
3. Explain how set point of body temp. is maintained. **(CMU:**Nov-23, Ja-08)
4. Mention the thermoregulatory response in human during exposure to cold. Write the pathophysiology of fever **(CMU:** May-22) (Ju-12).
5. Compare & contrast between hyperthermia & hyperpyrexia. (Ja-07)
6. Discuss the alarm and stress response of sympathetic nervous system. Name the sympathetic receptor with their blockers. (Ju-09)
7. Why sympathetic system is named as catabolic and parasympathetic system named as anabolic pathway? State the effects of sympathetic stimulation on eye. (Ja-07)
8. Contrast the sympathetic and parasympathetic system. Discuss in brief the changes that occur in fight or flight responses. (Ja-05, Ju-12)
9. Compare and contrast between the two divisions of autonomic nervous system. **(CMU:** Nov-22) Explain the role of sympathetic system in a state stress. (Ju-07, 10)
10. How temperature is regulated by hypothalamus? Thalamic syndrome (May-17)
11. **SN:** a) Heat stroke (Ju-09)
    b) Hypothermia (Ju-09)
    c) Asterognosis. (Ju-07)
    d) Thermoreceptor
    e) Frost bite (Ja-16)
12. What are the basic processes of heat loss from our body. (Ja-16)
13. What do you mean by core & shell temperature. (May-17)
14. Write down the function of thalamus. (May-17)
15. How temperature is gained and lost from our body. **(CMU:**May-21)
16. Mechanism of temparature regulation in cold climate. **(CMU:** Nov: 22, May-22)

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Structural Essay Questions (SEQ)

01. Define and discuss the basic steps of hemostasis. Name the substances that release from platelet and mention their importance. Explain the role of platelets in vascular spasm. **(CMU: May 2025)**

02. Define and classify blood pressure with normal values. List the mechanisms for short-term regulation of blood pressure. Describe the baroreceptor feedback mechanism for regulation of blood pressure. **(CMU: May 2025)**

03. Draw and label a neuronal control of gut wall. Give an account of electrical activity of GIT. How are they affected by sympathetic and parasympathetic stimulus? **(CMU: May 2025)**

04. Define ECG with its importance. Write the significance of different waves and intervals of ECG. Define heart block and ventricular escape.
(CMU: May 2025)

05. Discuss spermatogenesis with hormonal regulation. State the composition of semen. Write down the effect of sperm count and motility on fertility. **(CMU: Nov 2025)**

06. State the effects of insulin on adipose tissue, liver & muscle. Write down the mechanism of action of insulin with diagram. Give the flow chart of consequences of insulin deficiency in the body. **(CMU: Nov 2025)**

07. Compare the photoreceptor cells of retina. Mention the effect of visual pathway lesion with diagram. How does light adaptation occur? **(CMU: Nov 2025)**

08. Compare & contrast the antero-lateral and dorsal column pathway. Describe the origin, course with diagram, termination and functions of lateral spinothalamic tract. **(CMU: Nov 2025)**

09. Define and classify blood pressure. Mention the normal values of different types of pressure. Define hypertension. Write down the secondary causes of hypertension. Describe how blood pressure is regulated by long-term mechanism. **(CMU: May 2025)**

10. Define action potential. Describe the stages of a typical action potential with diagram. Explain how action potential is propagated along a large nerve fiber. State the saltatory conduction of action potential. **(CMU: May 2025)**

11. State the mechanism of O transport from lungs to tissue. Draw and explain O-Hb dissociation curve. Why is O-Hb dissociation curve shifted to the right during exercise? **(CMU: May 2025)**

12. Write down the composition and functions of gastric juice. Describe the mechanism of HCI secretion with diagram. Explain how pancreatic juice secretion is regulated**(CMU: May 2025)**

13. Define & classify hormone. Mention the name and importance of stress hormone & life-saving hormone. Explain the mechanism of action of steroid hormone. **(CMU: Nov 2025)**

14. Define endometrial cycle. Discuss the different phases of endometrial cycle. Write down the hormonal regulation of this cycle. **(CMU: Nov 2025)**

15. Define and classify reflex. Explain the two properties of reflex action. Write down the components of a reflex are with diagram. **(CMU: Nov 2025)**

16. List the common sites of recording temperature in human body. Define core and shell temperature. How is body temperature maintained in hot climate? What are the process of heat gain of the body? **(CMU: Nov 2025)**

17. Mention the indication of blood grouping What safety precautions are taken during blood transfusion? Why is cross-matching important? Write the hazards of mismatched blood transfusion. **(CMU: May-24)**

18. Define and classify blood pressure with their normal values. State the role of vasomotor center in regulation of blood pressure. Explain the role of kidney in blood pressure regulation. Define hypertension.**(CMU: May-24)**

19. Write the important constituents of saliva, gastric juice and pancreatic juice with their functions. How gastric juice secretion is regulated? Mention the factors that preventing acid-peptic digestion. **(CMU: May-24)**

20. Explain how 02 is transported from atmosphere to tissue cell & CO2 from tissue to atmosphere? Briefly discuss about Hering breuer reflex with its importance. What do you mean by Chyne-Stokes breathing?
(CMU: May-24)

21. Classify blood pressure with their normal values. Briefly discuss about blood pressure reeulatory mechanisms that activate within second. **(CMU: Nov-24)**

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22. How platelet plug is formed? Briefly discuss about prothrombin activator formation in intrinsic pathway. Name the bleeding disorders and mention the tests that should be done to diagnose a case of bleeding disorder. (CMU: Nov-24)

23. Draw and label oxyhemoglobin dissociation curve and explain its different parts. What is P50? Mention its significance. **(CMU: Nov-24)**

24. Name the principal digestive enzymes. Give their precursors, substrates and products of action of enzymes. Write down the fate of end product of fat digestion. **(CMU: Nov-24)**

25. Name the hormones secreted from adrenal glands. How is mineralocorticoid secretion regulated? Write down the physiological effects of glucocorticoids. **(CMU: May-24)**

26. Explain the mechanism of ovulation.
What is anovulatory cycle? Briefly discuss about the relation in between Ovarian cycle and uterine cycle. **(CMU: May-24)**

27. Enumerate the physiological divisions and functions Of cerebellum Explain how cerebral & cerebellum control distal limb movement? Mention the effects of cerebellar lesion. **(CMU: May-24)**

28. Draw and labe the visual pathway. State the visual impairment when any lesion in optic chiasma and left optic tract? Briefly discuss the changes occurs during accommodation response for near vision.**(CMU: May-24)**

29. Briefly discuss about thyroid hormone synthesis. Write down the features of myxoedema and cretinism. **(CMU: Nov-24)**

30. Define ovulation. Write down briefly about the indicators of ovulation. Give the flow chart of mechanism of ovulation. **(CMU: Nov-24)**

31. Define and classify deafness with causes. How will you differentiate different types of deafness. **(CMU: Nov-24)**

32. Draw and label different types of synapse. Briefly discuss about the properties of synapse. **(CMU: Nov-24)**

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